

William Abbruzzese

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Education

Colorado College, Bachelor's in Physics

May 2028

GPA: 3.96/4.0, Dean's List Fall 2024–Present

Relevant Coursework: Computational Physics, Mechanics, Observational Astronomy, Electronics, Techniques of Experimental Physics, Modern Physics, Applied Linear Algebra

Research & Engineering Projects

Autonomous UAV GNC Research | Colorado College | *Principal Investigator & Lead Engineer*

April 2026 – Present

- Awarded a \$5,000 institutional grant to design and build a search & rescue drone from the frame up, leading a team of 3 engineers and partnering with Multnomah County Sheriff's Office Search & Rescue.
- Wired and integrated avionics, GPS, telemetry, power, and propulsion electronics; built a custom MAVLink ground control station for interactive search-area planning and live telemetry.
- Designed terrain-optimized guidance algorithms reducing search time by 9%, integrating Bayesian search theory and Koopman kinematic models to dynamically update target zones using Probability of Detection metrics.
- Developed a three-tier natural language command parser translating voice into flight trajectories; validated via 25 autonomous flight tests with a 96% command-execution success rate.

Rocket Avionics & Thrust Vector Control | Independent Project

June 2025 – Present

- Designed a thrust-vector-control flight computer from scratch on a Teensy 4.1, fusing IMU and barometric data via a complementary filter to drive a two-axis gimbaled motor mount for active attitude stabilization
- Modeled and tuned the control system in MATLAB/Simulink and a custom Python simulation, and assessed propulsive-landing feasibility using real solid-motor thrust curves to quantify control-authority and throttle limits.
- Engineered the airframe and avionics in CAD (nose cone, electronics bay, gimbal) with custom power distribution and vibration isolation; earlier iterations built a fully analog PID controller from discrete components (without a microcontroller) to validate 2D stabilization on a self-balancing platform.

Optical Tracking of Artemis II Orion Spacecraft | Colorado College

April 2026

- Generated ephemerides through NASA's Horizons System and integrated them into TheSkyX to acquire and track Orion during its trans-Earth coast, resolving a non-cataloged target roughly 200,000 miles away.
- Captured a three-frame sequence resolving the capsule's motion against a fixed star field.

Micro-Wind Turbine Electromechanical Design | Independent Volunteer Project

June 2023 – Aug 2023

- Designed and manufactured a functional small-scale wind turbine from salvaged electromechanical components, implementing circuit fabrication and power distribution to deliver reliable lighting to a rural facility in Nicaragua lacking electrical infrastructure.

Leadership & Professional Experience

Team Leader | Multnomah County Search and Rescue – Portland, OR

Sept 2021 – Aug 2024

- Directed field operations for cross-functional teams of 10–15 individuals in high-pressure environments, emphasizing safety, communication, and rapid problem-solving.
- Logged 1,000+ operational hours as a 911 wilderness first responder locating and assisting missing or injured individuals in remote terrain throughout Oregon.
- Coordinated training and field operations as highest-ranked Team Leader, strengthening leadership, stakeholder coordination, and high-reliability teamwork.

Outdoor Gear Technician | Ahlberg Gear House – Colorado Springs, CO

March 2025 – Present

- Repair and calibrate technical outdoor equipment to safety-critical tolerances; advise customers on gear selection and trip planning, and manage inventory systems tracking thousands of items.

Technical Skills

- **Software:** Python, MATLAB/Simulink, Mathematica, C++ , SolidWorks, LTSpice, Altium, TheSkyX.
- **Embedded & Robotics:** ArduPilot, PX4, MAVLink, MAVProxy, Arduino, QGroundControl.
- **Fabrication:** Soldering, 3D printing, PCB assembly, hand tools, lab instrumentation.
- **Languages:** Italian, Spanish (Oregon State Seal of Bilingualism).